2022 exam : EM

Thursday, 27 October 2022 9:51 am

$$\theta = (p, \sigma_1^2, \sigma_2^2)$$

$$n = 50 \qquad n' = 50$$

$$\underline{p(X_1, \cdots X_{100}, Z_1, \cdots Z_{50} \mid \theta)}$$

$$= \prod_{i=1}^{50} p(X_i \mid Z_i, \theta)\, p(Z_i \mid \theta) \prod_{i=51}^{100} p(X_i \mid \theta)$$

$$= \prod_{i=1}^{50} \prod_{k=1}^{2} \left[p(X_i \mid Z_i = k, \theta)\, p(Z_i = k \mid \theta) \right]^{I\{Z_i = k\}} \prod_{i=51}^{100} p(X_i \mid \theta)$$

$$\boxed{\log p(\underline{\qquad\qquad\qquad})}$$

$$= \sum_{i=1}^{50} \Big\{ I(Z_i=1) \left[-\frac{1}{2}\log 2\pi\sigma_1^2 - \frac{x_i^2}{2\sigma_1^2} + \log p \right] + I(Z_i=2) \left[-\frac{1}{2}\log 2\pi\sigma_2^2 - \frac{x_i^2}{2\sigma_2^2} + \log(1-p) \right] \Big\}$$

$$+ \sum_{i=51}^{100} \left[-\frac{1}{2}\log 2\pi\sigma_1^2 - \frac{x_i^2}{2\sigma_1^2} \right]$$

(a)

$$\text{Exp} \boxed{\qquad\quad}$$

$$I(Z_i=1) \Rightarrow p(Z_i=1 \mid X_i, \theta^0) \overset{\text{let}}{=} A_{i1}$$

$$I(Z_i=2) \Rightarrow p(Z_i=2 \mid X_i, \theta^0) = A_{i2}$$

$$\frac{\partial Q(\theta, \theta^0)}{\partial \sigma_2^2} = \sum_{i=1}^{50} A_{i2} \left[-\frac{1}{2}\frac{1}{\sigma_2^2} + \frac{x_i^2}{2}\frac{1}{(\sigma_2^2)^2} \right]$$

$$= 0$$

$$\text{↯} \quad -\frac{1}{2(\sigma_2^2)^2} \left(\sigma_2^2 \sum_{i=1}^{50} A_{i2} - \sum_{i=1}^{50} A_{i2} x_i^2 \right) = 0$$

$$\boxed{\hat{\sigma}_2^2 = \frac{\sum_{i=1}^{50} A_{i2} x_i^2}{\sum_{i=1}^{50} A_{i2}}} \quad (b)$$

$$\frac{\partial Q(\theta, \theta^0)}{\partial \sigma_1^2} = \sum_{i=1}^{50} A_{i1} \left[-\frac{1}{2}\frac{1}{\sigma_1^2} + \frac{x_i^2}{2}\frac{1}{(\sigma_1^2)^2} \right]$$

$$+ \sum_{i=50}^{100} \left[-\frac{1}{2}\frac{1}{\sigma_1^2} + \frac{x_i^2}{2}\frac{1}{(\sigma_1^2)^2} \right] = 0$$

$$-\frac{1}{2(\sigma_1^2)^2} \Big(\sigma_1^2 \sum_{i=1}^{50} A_{i1} + \sigma_1^2\, 50 - \sum_{i=1}^{50} A_{i1} x_i^2 - \sum_{i=50}^{100} x_i^2 \Big) = 0$$

$$\boxed{\hat{\sigma}_1^2 = \frac{\sum_{i=1}^{50} A_{i1} x_i^2 + \sum_{i=50}^{100} x_i^2}{\sum_{i=1}^{50} A_{i1} + 50}} \quad (b)$$